ARDIAL

Software Engineer · TypeScript · Node.js · Go · Blockchain
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A lifelong learner, driven by curiosity long before the rise of AI, I have continuously expanded my knowledge across software engineering and emerging technologies. This mindset has shaped me into a **T-shaped Software Engineer**. My experience spans the **entire product lifecycle**, from system design and development to deployment, production operations, and continuous improvement. This end-to-end perspective allows me to understand not only how to build software, but also how engineering decisions impact reliability, cost, and the business.

EXPERIENCE

PT. Polaris Web
Jun 2026 – Present | Full-time · On-site

Software Developer

- Defined how the team builds software, setting a development pipeline of spec-driven development, test-driven development, AI-assisted workflows, automated deployment and security checks that is now followed by the entire division.
- Increased marketing team productivity by 30% by integrating AI-powered content generation into their article-writing workflow, reducing repetitive manual work and accelerating content production.
- Designed the Identity Provider (IdP) architecture for the CRM ecosystem, standardizing authentication and communication between modules while strengthening application-layer security and enabling secure cross-module interaction.
- Architected a Payment Orchestration Layer that abstracts integrations between applications and multiple Payment Service Providers (PSPs), providing a unified interface for payment operations, reducing integration complexity, and minimizing the engineering effort required to onboard and maintain payment providers.

Universitas IVET Semarang
Aug 2026 – Present | Part-time · WFA

Full-stack Engineer

Rebuilding Siveta end to end, working closely with stakeholders to define requirements and development specifications, designing the system architecture and UI/UX, and building the application across frontend and backend. Responsible for core academic features such as KRS and LMS, as well as database design, deployment, and CI/CD pipelines.

PT. Tekno Agri Karya
Aug 2025 – Jun 2026 | Hybrid

Full-Stack & Blockchain Engineer

- Developed and integrated multi-language SDKs (Golang, TypeScript, Rust) for web and backend service consumption, enabling seamless interaction with smart contract infrastructure.
- Built backend services and REST APIs supporting a Decentralized Identity (DID) platform, implementing a hybrid cryptographic data exchange layer (ECIES + symmetric encryption) for secure identity data handling.
- Optimized frontend performance by compiling Rust modules to WebAssembly (WASM), reducing client-side processing time by 50% without backend changes.
- Reduced storage usage by 90% and computation costs by 80% by designing a custom encoding/decoding layer for the internal HNS (Hara Name System).
- Contributed to smart contract development (ERC-4337 account abstraction with social recovery and gasless transactions) alongside the full web integration layer.

PT. Karya Kaya Bahagia
Jul 2025 – Present | Remote

Fullstack Developer Mentor Assistant

- Mentored 200+ students across 8 months (10–20 active students per week) through fullstack web development missions covering HTML/CSS, Node.js/Express, REST API design, ERD modeling, and layered architecture patterns.
- Assisted in assessing and refining the curriculum to align with industry best practices, and provided hands-on technical support that helped students meet deadlines and pass assessments.

PT. Gilang Maju Jaya
Jul 2023 – Feb 2024 | Remote · Part-time

Fullstack Blockchain Engineer

- Led fullstack development of ArcaneLabyrinth's game governance system, where players set the rules through on-chain proposals and voting, delivering 100% of the backend (NestJS), 50% of the frontend (Next.js), and the full smart contract implementation in Scrypto on the Radix blockchain.
- Played a key role in securing a Radix Foundation grant of $5,000 USD through technical delivery and proposal involvement.

(Ju) Boyoung

Mar 2020 – Jan 2025 | South Korea · Full-time

Machine Operator

Worked as a blue-collar machine operator in South Korea on demanding 12-hour shifts, six days a week. Despite the intensive schedule, continuously pursued technical certifications through Udemy, Coursera, and Radix while taking on domestic and international freelance blockchain projects on Radix.

PT. Java Digital Nusantara

Feb 2019 – Aug 2019 | On-site · Part-time

Jr. Network Engineer

- Deployed and configured network infrastructure for 50+ mid-to-large enterprise clients, including MikroTik routers with PPPoE, bandwidth management, and firewall policies, achieving a 100% client problem resolution rate.
- Managed IP addressing, subnetting, DHCP, and DNS, performed point-to-point link alignment, and carried out ongoing network security monitoring.

PT. Indosat

Jan 2019 – Feb 2019 | On-site

Tower Technician (Freelance)

- Installed and aligned antennas and radio equipment across 3 telecommunication towers.
- Configured IP addressing/subnetting and verified signal quality for stable connectivity.

Universitas Padjadjaran

Jan 2017 – Jul 2017 | On-site

IT Support Intern

- Troubleshot network and software issues across OSI Layers 1–4, including DHCP, DNS, IP subnetting, and physical infrastructure.
- Supported users with OS reinstallation, driver troubleshooting, and general technical issues.

PROJECTS

Simpul — *AI · Operations coordination* *Live*

Turns messages in busy group chats into action. AI labels each message, matches it to rules a manager wrote, and routes it to the right role, or to that role's backup, straight to their phone. Built with Express, TypeScript, PostgreSQL (Drizzle, pg-boss), Google Gemini and React.

Mnemo — *AI · Fullstack* *Live*

A solo-built AI note-taking application with RAG-powered chat that cites your own notes, semantic search with pgvector, and automatic event extraction. Built with FastAPI, LangChain, Google Gemini, PostgreSQL with pgvector, React 18, TypeScript, TailwindCSS and Docker.

Treecord — *Blockchain · Tree survival evidence*

Pays tree planters only for trees that survive. Each tree gets six check-ins over 36 months, each signed on a phone under the tree with GPS and time, and each pays the planter only if the tree is alive, with 40% held until month 36. Built with Solidity (Foundry) on Base, a TypeScript backend with PostgreSQL and an on-chain indexer and relayer, and React with viem and Privy.

Agaro-Vote — *Blockchain · Hackathon*

A full-stack digital voting system combining centralized services with a private blockchain, with weighted voting power through a synthetic staking reward mechanism. Responsible for 100% of the Solidity smart contract development, and built the Rust-based event listener service that keeps on-chain events in sync with the backend.

QSwap — *DeFi* *Live*

A personal DeFi platform for token swaps, Smart Order Routing (SOR) and liquidity pool creation on Radix and Ethereum. It implements the ERC-20 standard and the Uniswap V2 constant product ($x \cdot y = k$) AMM model for decentralized trading, deployed and live as an independent project.

Systems Programming Toolchain — *Rust · Compiler / VM*

A complete compiler toolchain built from scratch in Rust, with an assembler (assembly to machine code), a VM translator (VM bytecode to assembly) and a high-level compiler (Jack language to VM code). It covers lexical analysis, AST construction and code generation, producing programs that compile and run correctly.

EDUCATION

Universitas Terbuka *2026 – Present | Remote · Indonesia*
Bachelor of Mathematics

- Studying mathematics with a focus on cryptography and computation, including linear and discrete mathematics and the maths behind the systems.

Harisenin.com *Jun 2025 – Jul 2025 | Remote · Indonesia*
Full-Stack Developer Bootcamp

- Graduated with a perfect technical score of 100/100 in Batch 16, completed in 2 months against an expected 6-month timeline, the fastest graduation in the batch.
- Promoted to Mentor Assistant upon graduation, supporting 200+ students in frontend and backend missions.

SMK Bakti Ilham *Jun 2015 – Jun 2018 | Bandung, Indonesia*
Computer Networking · 91/100

- Achieved 98/100 on Ujikom (competency exam) covering Network Configuration and Server Administration.
- Placed 5th in the West Java MikroTik Olympiad for Vocational High Schools and was promoted to teach the MikroTik MTCNA intro course.
- Served as Student Council Leader.

CERTIFICATIONS

- Ethical Hacking and Penetration Testing (EHPT+) — Udemy [Ocsaly], 2025
- Red Teaming: Exploit Development with Assembly and C (MSAC+) — Udemy [Ocsaly], 2025
- Full-Stack Developer Bootcamp — Harisenin.com, 2025
- C# .NET with MS SQL — Udemy [Dominic Tripodi], 2025
- Scrypto 101 — Radix Foundation, 2024 (won a $200+ challenge with an Escrow blueprint)
- Build Responsive Real-World Websites with HTML and CSS — Udemy [Jonas Schmedtmann], 2022
- Become an Algebra Master / Linear Algebra Master / Geometry Master / Master the Fundamentals of Math — Udemy [Krista King], 2022
- Problem Solving (Basic) — HackerRank, 2022
- MikroTik MTCNA — SMK Bakti Ilham, 2017 & 2018

LANGUAGES

Indonesian (native) | English (intermediate) | Korean, EPS-TOPIK (basic) | German, A2 (basic)

SKILLS

Frontend	React, Next.js, HTML, CSS, TypeScript, JavaScript
Backend	Node.js, NestJS, Golang, PHP/Laravel 12, Python, FastAPI, REST API, WebSocket
AI / LLM	LangChain, RAG, Agentic AI Systems, AI-assisted development pipelines
Engineering practice	Spec-driven development, Test-driven development (TDD), automated deployment, security checks
Blockchain	Rust, Solidity, Scrypto, WASM, Smart Contract Development
Infra & DevOps	Docker, Kubernetes, AWS (ECR, ECS, EC2, S3), GCP (GCE, ALB, Cloud Armor), Azure DevOps, CI/CD (GitHub Actions), Linux (Debian)
Databases	PostgreSQL, MySQL, MongoDB, Redis